

knowing DNA's structure, the study of heredity could not advance. By the 1950s, key players in the race were US chemist Linus Pauling, and Rosalind Franklin and Maurice Wilkins at King's College London.

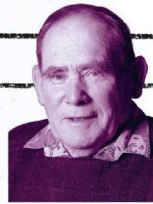
Scientists knew that DNA could pass on genetic information and make copies of itself. They also knew that the molecule was made up of various components: a "backbone" composed alternately of phosphoric acid and sugar, and four chemical "bases" called adenine (A), thymine (T), guanine (G), and cytosine (C). What was not clear was how these components fit together or how the molecule could replicate.

Dynamic duo

Crick and Watson applied themselves to the investigation with single-minded focus. They compiled evidence from other scientists and pooled their own combined knowledge of disciplines including viral genetics, physics, and X-ray crystallography to interpret the

SYDNEY BRENNER

South African biologist Sydney Brenner was one of the first to see Watson and Crick's DNA model.



After completing a PhD at Oxford University, Brenner (1927–2019) produced leading research into developmental genetics and molecular biology. In 1961, he discovered the encoding process for the amino acids of a protein with Crick. Brenner is best known for his work on organ development and cell apoptosis (programmed cell death) through his study of transparent nematodes, for which he jointly received a Nobel Prize in 2002.

data. In a break from convention, instead of conducting exhaustive experiments, they built large 3D models to try and replicate DNA's structure according to its known chemical properties. After two failed attempts, their success was made possible after obtaining data belonging to Rosalind Franklin (see pp.270–273).

“We have discovered the secret of life.”

Francis Crick, 1953

**MADE THEIR
DNA DISCOVERY
AFTER ONLY**

**18
MONTHS
OF RESEARCH**

**BUILT THEIR
FINAL
DNA
MODEL IN
3 DAYS**


**PUBLISHED THEIR
GROUND-
BREAKING
WORK
IN A PAPER
JUST 1
PAGE LONG**

